VTD 425: Human Technological Systems

University of Idaho Spring 2026 3 Credits

- **CRN:** 42918
- **Meeting:** Tuesdays & Thursdays, 2:00 - 3:15 PM
- **Dates:** Jan 15, 2026 - May 7, 2026
- **Location:** AAN Crit Space (In-person only)
- **Instructor:** Roger Lew
- **Contact:** rogerlew@uidaho.edu, 208-660-4525 (business hours)
- **Office Hours:** By appointment

No class March 17 & 19, 2026 (Spring Recess)

Course Description

Overview of user interaction design and evaluation of physical and virtual human-machine systems. We want to train the next generation of virtual integrated designers!

VTD is inherently interdisciplinary/transdisciplinary. We utilize knowledge and methods from various domains to:

- Understand the role of humans in highly sophisticated human-coupled systems
- Apply systems-thinking and first-principles to design solutions
- Model, simulate, and iterate to optimize designs
- Develop, deploy, and validate virtual technologies

Topics include introduction to human factors, human technological systems, usability evaluation, HCI in virtual reality, human error, human performance, mixed-initiative systems (intelligent and autonomous teaming), design and prototyping.

Recommended Preparation: Graduate student status (5xx) *or* undergraduate who has taken a VTD 300-Level production course and one VTD 400-Level history and theory course, or with instructor approval.

This is a 16-week in-person course to introduce Virtual Technology and Design students to human factors, human computer interaction design and evaluation, emphasizing mixed-initiative systems and virtual reality. The first 8-weeks are to develop the theory and during the last 8-weeks students driven topical discussions

About This Course

- This is a seminar course. Active participation is required. The idea is to not only familiarize students with the topics but to get students to relate to the topics.

- The first weeks provide background regarding Human Factors/ Cognitive Psychology/ Sensation and Perception/ Usability & User Experience. The content and discussion is oriented towards designers, engineers, and technologists.
- After that we explore contemporary technology topics that the students find relevant and the instructor deems interesting. This could include: self-driving cars, human enhancement/posthumanism, AI art, etc. The intent is to get to an academic level of discussion supported by papers and other in-depth materials. We generally plan a few weeks ahead and bounce around from topic to topic.

Assignments/Projects

1. Product Evaluation Review a product/application. Describe the good/bad using human factors concepts. Describe how the design could be improved.

Ideas to review: Smart Home Devices, Health and Fitness Apps, Mobile Payment Apps, Productivity Apps, Educational Technology, Augmented Reality Apps, Wearable Technology, Virtual Reality Platforms, Smart Assistants, Mobile Photography and Editing Apps, Environmental and Sustainability Apps, Language Learning Apps, Physical Products like Cars, Radios, Watches, ...

Document up to 5 pages and PowerPoint 4-5 slides with 5-minute presentation.

Due Thursday, February 19, 2026 (Presentation in class, Submit document to Canvas)

2. Technology Op-Ed Lightning Talk Take a position on a contemporary human-technology issue. Present a short op-ed style argument (5-10 minutes) to the class.

Topics could include: AI ethics, social media effects, screen time, autonomous vehicles, privacy, algorithmic bias, digital accessibility, technology addiction, workplace automation, surveillance, etc.

Focus on making a clear argument supported by evidence and human factors principles.

Due Tuesday, March 3 & Thursday, March 5, 2026 (Lightning talks in class)

3. AI Assisted Workflow Exploration Use an AI technology (e.g., ChatGPT, generative AI) to make something. Present what you made and how you made it. Describe the strengths and weaknesses of AI-teaming.

Focus on the process, not the product.

PowerPoint 4-5 slides with 5-minute presentation.

Due Thursday, April 16, 2026 (Presentation in class, Submit to Canvas)

4. Human Technological Systems Artifact Reflect on topics and ideas covered throughout the course. Select an idea and create an artifact that illustrates the concept or a point of view regarding the topic.

Artifact *noun* an object made by a human being, typically an item of cultural or historical interest.

Potential artifacts: - Infographic - Poem - Flyer - PowerPoint presentation - Video - Short essay - Comic - Website - Performance art

Objectives: - Review course material - Organize and synthesize ideas on a topic - Create a deliverable that illustrates the concept or your thoughts on the topic - Build something for your portfolio - Have something to discuss in future jobs/interviews

Due Saturday, March 7, 2026 (Submit to Canvas)

Tentative Schedule

Week 0 - Introduction to the Course, Introduction to Human Factors Psychology

Introductions

Course Description

What is Virtual Technology and Design?

What is Human Factors?

What is Science? (Philosophy of Science)

Research Design and Methods

Potential Topic Identification

Week 1,2 - Introduction to Human Factors

What is human factors? Principles of human factors

Materials:

YouTube: CASABriefing History of human factors

Donald A. Norman (2013). The Design of Everyday Things. Basic Books. Chapter 1. The PsychoPathology of Everyday Things.

Frank Buschmann (2010). Learning from Failure, Part 2: Featuritis, Performatis, and Other Diseases

Bordens & Abbott. Research Methods. Chapter 1. Explaining Behavior. (2024). PDF

Bill Gates Episode 6: Sam Altman https://www.youtube.com/watch?v=PkXELH6Y2lM

Week 3,4 - Usability, User Experience

Materials:

Usability Geek. The Difference Between Usability and User Experience.

NN/g. 10 Usability Heuristics

NN/g. 10 UX Research Cheat Sheet

GoDaddy: Web Accessibility Guidelines in About 7 Minutes

Universal Design

Potential Topics for Discussion

After the first 4 weeks of directed content, the class transitions to student-driven topical explorations of contemporary technologies. The following are potential discussion topics—students are encouraged to propose additional topics of interest.

Artificial Intelligence

- **AI as UI Paradigm** - NN/g: AI: First New UI Paradigm in 60 Years
- **Large Language Models** - Microsoft Research: Sparks of AGI
- **Generative AI** - Cold Fusion: It's Time to Pay Attention to A.I.
- **AI Companions** - The rise of AI Companion
- **AI/Human teaming** - body doubling, accessibility accommodations for neurodivergence

Spatial Computing & XR

- **Mixed Reality** - HYPER-REALITY
- **Apple Vision Pro** - Casey Neistat, MKBHD
- **Metaverse** - Work in the metaverse
- **VR/AR applications** - environmental accommodations, transfer training
- **Tracking and gesturing technologies**

Robotics & Automation

- Humanoid robots
- Empathetic robots and AI
- Hospitality applications (hotels, airports)
- Autonomous systems displacing human labor

Human Enhancement

- Brain computer interfaces (Neuralink, etc.)
- Gene therapy and personalized medicine (mRNA)

- AI personal assistance
- Designer babies and genetic selection
- Transhumanism vs. posthumanism
- Human nature, dignity, and immortality

Emerging Technologies

- **Nanotechnology and swarms** - medical applications, enhancement
- **Decentralization** - distributed networks, energy, information
- **Healthcare technology**
- **Transfer training and simulation**

Grading

Component	Weight
Participation/Attendance	40%
1. Product Evaluation	15%
2. Technology Op-Ed Lightning Talk	15%
3. AI Assisted Workflow	15%
4. Human Technological Systems Artifact	15%
Total	**100%**

Grading Scale: A = 90-100%, B = 80-89%, C = 70-79%, D = 60-69%, F = <60%

Course Policies & Procedures

Attendance Policy

Regular class attendance and participation is expected. For a seminar like this your attendance should be 100%. Please let me know in advance when you will miss a class for any reason. More than one absence will lead to a reduction of your participation grade.

This is a seminar course. You are expected to read the assigned materials before class and be prepared to discuss the information during class. Your constructive and informed participation is essential. Preparedness factors into the attendance/participation grade.

Late Assignments

Unless otherwise instructed, please submit your assignments through Canvas. Late assignments are not accepted without an approved extension before the due date.

Artificial Intelligence Policy

Use of AI is permitted and encouraged. However, users of AI are ultimately responsible for the work. Use AI responsibly and please disclose AI-assisted content.

Learning Outcomes

This course aligns with the University of Idaho's learning outcomes:

1. **Think and create** - Use multiple strategies to examine real-world issues, explore creative avenues of expression, solve problems, and make consequential decisions.
2. **Communicate** - Acquire, articulate, create and convey intended meaning using verbal and non-verbal methods of communication that demonstrate respect and understanding in a complex society.
3. **Practice citizenship** - Apply principles of ethical leadership, collaborative engagement, socially responsible behavior, respect for diversity in an interdependent world, and a service-oriented commitment to advance and sustain local and global communities.

Academic Integrity

The University of Idaho expects all students to adhere to the highest standards of academic honesty. Please review the UI Student Code of Conduct Article II - Academic Honesty.

Plagiarism is using someone else's words, ideas, or work and presenting it as your own. Your assignments should be your original work. When using outside sources, you must cite them properly.

Learning Civility

In any environment in which people gather to learn, it is essential that all members feel as free and safe as possible in their participation. Everyone in this course will be treated with mutual respect and civility.

Center for Disability Access & Resources (CDAR)

UI is committed to ensuring an accessible learning environment. If you believe that you qualify for disability-related academic adjustments, please contact CDAR to discuss eligibility. Disability-related academic adjustments are not retroactive.

- **Location:** Bruce Pitman Building, Suite 127
- **Phone:** 208-885-6307
- **Email:** cdar@uidaho.edu
- **Website:** www.uidaho.edu/current-students/cdar

Nondiscrimination Policy

The University of Idaho prohibits discrimination on the basis of race, color, religion, national origin, sex, age, disability, or veteran status. Complaints about discrimination or harassment should be brought to the attention of the Office of Civil Rights and Investigations (208-885-4285).

UI Moscow Land Acknowledgement

The University of Idaho welcomes and respects all people. UI Moscow is located on the homelands of the Nimiipuu (Nez Perce), Palus (Palouse), and Schitsu'umsh (Coeur d'Alene) tribes. We extend gratitude to the indigenous people that call this place home, since time immemorial.

University of Idaho Supports

Academic Supports

Resource	Contact	Location
Academic Coaching	acadcoaching@uidaho.edu	ISUB, 3rd Floor
Career Services	208-885-6121	ISUB, 3rd Floor
CDAR	208-885-6307	Bruce M. Pitman Center, Suite 127
Library	208-885-6534	850 S Rayburn Street
Writing Center	208-885-6644	Library, 2nd Floor, Room 215

Health and Wellness

Resource	Contact	Location
Counseling & Mental Health	208-885-6716	Mary Forney Hall, Room 306
VandalCARE	208-885-6757	TLC, Room 232
Vandal Health Clinic	208-885-6693	831 Ash Street

Emergency and Safety

Resource	Contact
Moscow Police Sub-Station	208-882-2677
Safe Walk (24/7)	208-885-SAFE (7233)
Security Services	208-885-7054

VandalCARE: If you are concerned about a fellow Vandal, submit a VandalCARE referral. For emergencies, call 911 immediately.

Everything else: see Canvas